EDS 223 Homework 3: West Coast Aquaculture Analysis

Kate Frankland

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Table of contents

Setup 1
Find Suitable Locations 4
Oyster Map Creation 7
Pacific Calico Scallop 9
Scallop Map Creation 12

Setup

```
# Loading packages
library(terra) # raster handling
library(tidyverse)
library(tmap) # map making
library(kableExtra) # table formatting
library(spData) # spatial data
library(spDataLarge) # spatial data
library(geodata) # spatial data
library(sf)
```

```
# Shapefile for the West Coast EEZ

ez_data <- st_read("data/wc_regions_clean.shp")
```

```
Reading layer `wc_regions_clean' from data source
```

```
  `/home/katefrankland/west_coast_aquaculture_analysis/data/wc_regions_clean.shp'
  using driver `ESRI Shapefile'
Simple feature collection with 5 features and 5 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -129.1635 ymin: 30.542 xmax: -117.097 ymax: 49.00031
Geodetic CRS:  WGS 84
```

```
# Bathymetry Raster

bathymetry <- rast("data/depth.tif")
```

```
#SST rasters (combine SSt rasters into a raster stack)

ann_avg_sst_2008 <- rast("data/average_annual_sst_2008.tif")
ann_avg_sst_2009 <- rast("data/average_annual_sst_2009.tif")
ann_avg_sst_2010 <- rast("data/average_annual_sst_2010.tif")
ann_avg_sst_2011 <- rast("data/average_annual_sst_2011.tif")
ann_avg_sst_2012 <- rast("data/average_annual_sst_2012.tif")

# test class of raster objects
class(ann_avg_sst_2008)
```

```
[1] "SpatRaster"
attr(,"package")
[1] "terra"
```

```
# Combine rasters into a stack
raster_stack <- c(
    ann_avg_sst_2008,
    ann_avg_sst_2009,
    ann_avg_sst_2010,
    ann_avg_sst_2011,
    ann_avg_sst_2012)
```

```
#Checking the CRS
st_crs(ez_data) == st_crs(bathymetry)
```

```
[1] TRUE
```

```r
st_crs(ez_data) == st_crs(raster_stack)
```

```
[1] TRUE
```

```r
st_crs(bathymetry) == st_crs(raster_stack)
```

```
[1] TRUE
```

##Process data Next, we need to process the SST and depth data so that they can be combined. In this case the SST and depth data have slightly different resolutions, extents, and positions.

```r
# find the mean SST from 2008-2012 (e.g. create single raster of average SST)

sst_mean <- mean(raster_stack, na.rm = TRUE)
print(sst_mean)
```

```
class       : SpatRaster
size        : 480, 408, 1  (nrow, ncol, nlyr)
resolution  : 0.04166185, 0.04165702  (x, y)
extent      : -131.9848, -114.9867, 29.99305, 49.98842  (xmin, xmax, ymin, ymax)
coord. ref. : lon/lat WGS 84
source(s)   : memory
name        :    mean
min value   :  278.13
max value   : 306.045
```

```r
# convert average SST from Kelvin to Celsius hint: subtract by 273.15
sst_celsius <- (sst_mean - 273.15)
print(sst_celsius)
```

```
class       : SpatRaster
size        : 480, 408, 1  (nrow, ncol, nlyr)
resolution  : 0.04166185, 0.04165702  (x, y)
extent      : -131.9848, -114.9867, 29.99305, 49.98842  (xmin, xmax, ymin, ymax)
coord. ref. : lon/lat WGS 84
source(s)   : memory
name        :   mean
min value   :   4.98
max value   : 32.895
```

```r
# crop depth raster to match the extent of the SST raster
cropped_depth <- crop(bathymetry, sst_celsius)
```

```r
# note: the resolutions of the SST and depth data do not match resample the depth data to ma
```

```r
resampled_depth <- resample(cropped_depth, sst_celsius, method = "bilinear")

# Vverify matching
cat("After resampling:\n")
```

```
After resampling:
```

```r
cat("Resolutions match:", all(res(resampled_depth) == res(sst_celsius)), "\n")
```

```
Resolutions match: TRUE
```

```r
cat("Origins match:    ", all(round(origin(resampled_depth), 8) == round(origin(sst_celsius)
```

```
Origins match:     TRUE
```

```r
cat("Extents match:    ", ext(resampled_depth) == ext(sst_celsius), "\n")
```

```
Extents match:     TRUE
```

Find Suitable Locations

- To find suitable locations for marine aquaculture, we'll need to find locations that are suitable in terms of both SST and depth.
- reclassify SST and depth data into locations that are suitable for oysters
- hint: set suitable values to 1 and unsuitable values to 0
- find locations that satisfy both SST and depth conditions

```r
# Define a reclassification matrix
sst_rcl <- matrix(c(-Inf, 11, 0, # unsuitable
                    11, 30, 1,  # suitable
                    30, Inf, 0), # unsuitable
                  ncol = 3,
                  byrow = TRUE)

# Apply the matrix to reclassify the raster, making all cells 0 or 1 or 2
sst_suitability <- classify(sst_celsius, rcl = sst_rcl, right = FALSE)
```

```r
# Reclassify depth for oysters

depth_rcl <- matrix(
  c(
    -Inf, -70, 0,  # Unsuitable
    -70, 0, 1,     # Suitable
    0, Inf, 0      # Unsuitable
  ),
  ncol = 3,
  byrow = TRUE
)

# Project and resample depth to match SST raster
depth_projected <- project(
  resampled_depth,
  sst_celsius,
  method = "bilinear"
)

# Reclassify projected depth raster
depth_suitability <- classify(
  depth_projected,
  rcl = depth_rcl,
  right = FALSE
)

# Check that CRS matches
same.crs(sst_suitability, depth_suitability)
```

```
[1] TRUE
```

```r
# Combine the rasters
suitable_zones <- c(
  sst_suitability,
  depth_suitability
)

# Multiply suitable zones to find suitable locations
suitable_locations <- lapp(
  suitable_zones,
  fun = function(sst, depth) sst * depth
)
```

##Determine the most suitable EEZ - We want to determine the total suitable area within each EEZ in order to rank zones by priority. To do so, we need to find the total area of suitable locations within each EEZ.

```r
# select suitable cells within West Coast EEZs
eez_rast <- rasterize(ez_data, suitable_locations, field = "rgn_id")

# find area of grid cells
cell_size <- cellSize(suitable_locations, unit = "km")

# Only want the suitable cells
suitable_area <- suitable_locations * cell_size

# find the total suitable area within each EEZ
# hint: it might be helpful to rasterize the EEZ data

# Calculate suitable area by EEZ
eez_suitable <- zonal(
  suitable_area,
  eez_rast,
  fun = "sum",
  na.rm = TRUE
)

# Rename suitable area column
eez_suitable <- eez_suitable |>
  rename(
    suitable_km2 = lyr1
  )
```

```r
# Join suitable area to EEZ polygons
ez_summary <- ez_data |>
  left_join(eez_suitable, by = "rgn_id")

# Check results
head(ez_summary)
```

```
Simple feature collection with 5 features and 6 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -129.1635 ymin: 30.542 xmax: -117.097 ymax: 49.00031
Geodetic CRS:  WGS 84
                  rgn rgn_key      area_m2 rgn_id  area_km2 suitable_km2
1              Oregon      OR 179994061293      1 179994.06    1395.7328
2 Northern California    CA-N 164378809215      2 164378.81     323.9124
3  Central California    CA-C 202738329147      3 202738.33    4373.5565
4 Southern California    CA-S 206860777840      4 206860.78    3329.9002
5          Washington      WA  66898309678      5  66898.31    3091.5158
                        geometry
1 MULTIPOLYGON (((-123.4318 4...
2 MULTIPOLYGON (((-124.2102 4...
3 MULTIPOLYGON (((-122.9928 3...
4 MULTIPOLYGON (((-120.6505 3...
5 MULTIPOLYGON (((-122.7675 4...
```

Oyster Map Creation

```r
# Map creation

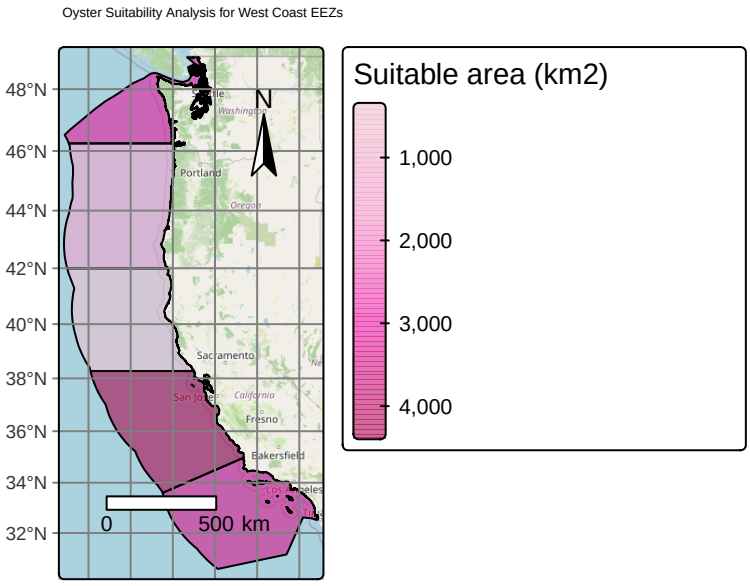
#Create color values

pink_theme <- c("#eec0cd", "#f39bcc", "#e81ea4", "#a9074d")
```

```r
oyster_map <- tm_basemap("OpenStreetMap") +
  tm_shape(ez_summary) +
  tm_polygons(
    fill = "suitable_km2",
    fill.scale = tm_scale_continuous(values = pink_theme),
    fill_alpha = 0.6,
    fill.legend = tm_legend(
```

```
      title = "Suitable area (km2)",
      position = tm_pos_out("right", "center"),
      width = 15,
      height = 15
    ),
    col = "black"
  ) +
  tm_graticules() +
  tm_scalebar(position = c("left", "bottom"), text.size = 0.7) +
  tm_compass(position = c("right", "top")) +
  tm_options(components.autoscale = FALSE) +
  tm_title("Oyster Suitability Analysis for West Coast EEZs", size = 0.5)

# Display map
oyster_map
```



Oyster Map Interpretation: The oyster map shows suitable habitat across the West Coast EEZs, primarily Southern California and parts of Washington State. Based on the selected criteria, these regions have the most potential for oyster aquaculture success.

#_____

Pacific Calico Scallop

link: https://www.sealifebase.ca/Country/CountrySpeciesSummary.php?c_code=591&Genus=Argopecten&Species=ventricosus

```
# Suitability for the Pacific Calico Scallop

# Suitability parameters
# Depth range: 0-180 meters below sea level

# Reclassify SST for the CA mussel

scallop_sst_rcl <- matrix(
  c(
    -Inf, -21, 0,       # Unsuitable
    21, 31, 1,          # Suitable
    31, Inf, 0          # Unsuitable
  ),
  ncol = 3,
  byrow = TRUE
)

scallop_sst_suitability <- classify(
  sst_celsius,
  rcl = scallop_sst_rcl,
  right = FALSE
)

# Project and resample depth to match SST raster

scallop_depth_projected <- project(
  resampled_depth,
  sst_celsius,
  method = "bilinear"
)

# Reclassify depth for Pacific calico scallop

scallop_depth_rcl <- matrix(
  c(
    -Inf, -180, 0,     # Unsuitable
    -180, 0, 1,        # Suitable
    0, Inf, 0          # Unsuitable
```

```
  ),
  ncol = 3,
  byrow = TRUE
)

scallop_depth_suitability <- classify(
  scallop_depth_projected,
  rcl = scallop_depth_rcl,
  right = FALSE
)

# Check that CRS matches

same.crs(
  scallop_sst_suitability,
  scallop_depth_suitability
)
```

```
[1] TRUE
```

```
# Combine the rasters

scallop_suitable_zones <- c(
  scallop_sst_suitability,
  scallop_depth_suitability
)

# Multiply suitable zones to find suitable locations

scallop_suitable_locations <- lapp(
  scallop_suitable_zones,
  fun = function(sst, depth) sst * depth
)


# Select suitable cells within West Coast EEZs

scallop_eez_rast <- rasterize(
  ez_data,
  scallop_suitable_locations,
  field = "rgn_id"
)
```

```r
# Find area of grid cells

scallop_cell_size <- cellSize(
  scallop_suitable_locations,
  unit = "km"
)

# Only want the suitable cells

scallop_suitable_area <-
  scallop_suitable_locations * scallop_cell_size

# Calculate suitable area by EEZ

scallop_eez_suitable <- zonal(
  scallop_suitable_area,
  scallop_eez_rast,
  fun = "sum",
  na.rm = TRUE
)

# Rename suitable area column

scallop_eez_suitable <- scallop_eez_suitable |>
  rename(
    suitable_km2 = lyr1
  )

# Join suitable area to EEZ polygons

scallop_summary <- ez_data |>
  left_join(scallop_eez_suitable, by = "rgn_id")

# Check results

head(scallop_summary)
```

```
Simple feature collection with 5 features and 6 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -129.1635 ymin: 30.542 xmax: -117.097 ymax: 49.00031
```

```
Geodetic CRS:  WGS 84
                  rgn rgn_key      area_m2 rgn_id  area_km2 suitable_km2
1              Oregon      OR 179994061293      1 179994.06    177472.94
2 Northern California    CA-N 164378809215      2 164378.81     82176.73
3  Central California    CA-C 202738329147      3 202738.33    112935.09
4 Southern California    CA-S 206860777840      4 206860.78    121965.65
5          Washington      WA  66898309678      5  66898.31    174775.80
                        geometry
1 MULTIPOLYGON (((-123.4318 4...
2 MULTIPOLYGON (((-124.2102 4...
3 MULTIPOLYGON (((-122.9928 3...
4 MULTIPOLYGON (((-120.6505 3...
5 MULTIPOLYGON (((-122.7675 4...
```

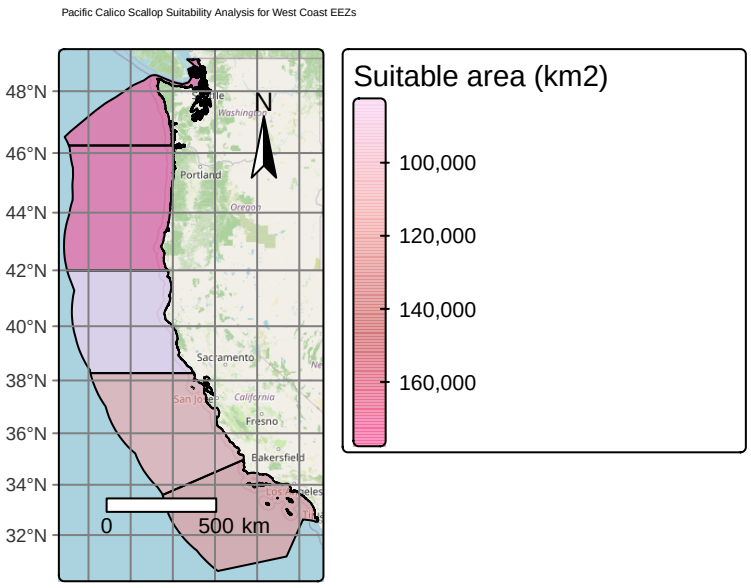
Scallop Map Creation

```r
# Scallop color theme

scallop_theme <- c("#f9d0f1", "#f6a8a8", "#cc5f5f", "#f9529a")

scallop_map <- tm_basemap("OpenStreetMap") +
  tm_shape(scallop_summary) +
  tm_polygons(
    fill = "suitable_km2",
    fill.scale = tm_scale_continuous(values = scallop_theme),
    fill_alpha = 0.6,
    fill.legend = tm_legend(
      title = "Suitable area (km2)",
      position = tm_pos_out("right", "center"),
      width = 15,
      height = 15
    ),
    col = "black"
  ) +
  tm_graticules() +
  tm_scalebar(position = c("left", "bottom"), text.size = 0.7) +
  tm_compass(position = c("right", "top")) +
  tm_options(components.autoscale = FALSE) +
  tm_title("Pacific Calico Scallop Suitability Analysis for West Coast EEZs",
    size = 0.5
  )
```

```
# Display map
scallop_map
```



Scallop map interpretation: The Pacific Calico Scallop Suitability Map shows that there is suitable habitat across the West Coast EEZs based on the selected temperature and depth criteria. Southern California, Oregon, and parts of Washington state have habitat that meet the criteria for this scallop.